

PAPR reduction techniques optimization-based OFDM signal for wireless communication systems

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ABSTRACT

OFDM (Orthogonal Frequency Division Multiplexing) is a widely used modulation technique in many standards such as Wi-Fi (Wireless Fidelity), LTE (Long Term Evolution), and 4 G. It allows for high transmission data rates and excellent spectral efficiency. However, one of the main limitations of OFDM is its high PAPR (Peak-to-Average Power Ratio). In this paper, we will first describe some methods to reduce the PAPR associated with OFDM signal. We will discuss some methods such as the clipping, the Selective mapping SLM, the Partial Transmit Sequence (PTS), and the Tone Reservation (TR). Let's list their advantages and disadvantages before focusing on the TR method which we will detail. This work aims to improve the performance of the TR method in terms of speed of convergence and reduction of the PAPR, using the conjugate different algorithms while respecting the frequency specifications of the IEEE 802.11a standard. The proposed algorithm which is based on the gradient method, Conjugate gradient methods, and Quasi-Newton method shows good performance regarding PAPR reduction and speed of convergence. The choice of an appropriate technique depends on several factors: required PAPR reduction, acceptable complexity, and performances (spectral efficiency, latency, etc.). A trade-off must be found in practice between these different criteria. It has been shown that the Quasi-Newton method algorithm with its two variants converges more quickly.

1. Introduction

Multi-carrier transmission, commonly referred to as orthogonal frequency division multiplexing (OFDM) is one of the most known modulation techniques that has been widely employed in wireless communication systems, due to its ability to transmit high-speed data with relatively large bandwidth and robust immunity to multipath fading. Multi-carrier communication systems compete with single-carrier systems, but they suffer from a serious disadvantage, namely a high peak-to-average power ratio (PAPR) due to the highly fluctuating envelope of transmitted signals [1]. For this purpose in this contribution, all the concepts relating to the problem of PAPR will be defined thereafter the methods for the reduction of this last will be described. These strong fluctuations of power pose several problems for the electronic components located downstream such as the amplifiers or ADC Analog-to-Digital Converters. They must manage both the high peak power of the peaks and the low average power, which induces high dynamics and performance degradation. Many techniques have been proposed in the literature, acting in different ways on the signal

(Clipping, Selective mapping SLM, Partial Transmit Sequence (PTS), Clipping, tone reservation, etc.) [2].

After explaining the different PAPR reduction methods, we have chosen to work with the TR (Tone Reservation) method, belonging to the class known as "signal addition". This choice was made taking into account its advantages, in particular the backward compatibility criterion. This means that the implementation of the method of transmission does not generate any modification on the receiver. For this, we will propose to optimize the TR method [3] in the speed of convergence and reduction of the PAPR with the use of the conjugate gradient algorithm. The impact of PAPR reduction with the TR method is generally studied by simulation in the presence of simple models of the amplifier, without considering phase distortions or even memory effects. Moreover, we try to prove that choosing the TR (Tone Reservation) technique is the best among the others in terms of spectral efficiency [4].

In this paper, the TR technique is used for OFDM signals. The presented method is compared with the Clipping, PTS, and SLM techniques in terms of PAPR reduction performance. The obtained results indicate that the proposed TR approach is more efficient than the Clipping, PTS,

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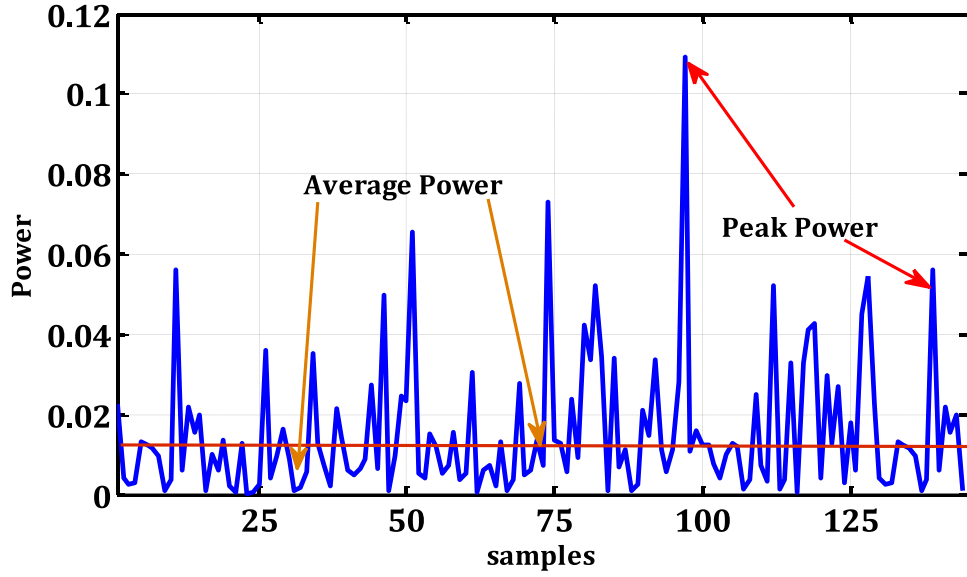


Fig. 1. Instantaneous power of the OFDM signal with cyclic prefix.

and SLM schemes. However, the Tone Reservation method has certain drawbacks. Firstly, it requires prior knowledge of the signals used in the system, especially the signal sequences that result in high PAPR. This can be challenging to obtain in some cases, making the application of the method more complex. Here, we used the Tone Reservation method and optimized it with some algorithms such as the gradient method, Conjugate gradient methods, and Quasi-Newton method to improve the performance of wireless communication systems by reducing non-linear distortion, improving signal quality, and optimizing energy efficiency. The structure of this contribution will be as follows: definition of PAPR, presentation of existing techniques to reduce the PAPR, detailed description of the proposed method, simulation results and comparison with the state of the art, optimization of TR approach, conclusion, and some perspectives.

2. Definition of papr and CCDF

2.1. Definition of PAPR

The problem of high Peak-to-Average Power Ratio (PAPR) is one of the major challenges encountered in Orthogonal Frequency Division Multiplexing (OFDM) systems. OFDM is a modulation technique widely used in wireless communications, including standards such as Wi-Fi, 4 G, and 5 G. OFDM divides the wideband signal into narrow subcarriers, which allows for better spectral efficiency and improved resistance to multi-path propagation effects. However, this division into subcarriers creates power peaks in the signal, resulting in a high PAPR. PAPR is defined as the ratio between the peak instantaneous power of the OFDM signal and its average power. High PAPR can cause several problems in communication systems:

High-power amplifier: Power amplifiers used in communication systems have limits in terms of the input power they can handle. High PAPR can cause nonlinear distortions in amplifiers, leading to degradation in signal quality and increased transmission errors.

Energy efficiency: A high PAPR means that the peak power of the signal is much higher than its average power [5]. This requires a more expensive amplifier design, as it needs to handle high-power peaks. This reduces the overall energy efficiency of the system. Analog-to-digital converter (ADC) dynamics, OFDM systems require analog-to-digital conversion to digitize the signal. High PAPR can cause saturation in the analog-to-digital converter, resulting in signal distortion and decreased signal recovery quality. Interference between subcarriers,

When PAPR is high, power peaks can cause interference between neighboring subcarriers. This can lead to signal distortion and increased demodulation errors.

Peak to Average Power ratio is the ratio between the maximum power and the average power of the OFDM signal over a time interval T (Fig. 1). The PAPR expressed in dB, is given by:

$$\text{PAPR}(x(t)) = \frac{P_{\max}}{P_{\text{Average}}} = \frac{\max(|x(t)|^2)_{0 \leq t \leq T}}{\frac{1}{T} \int_0^T |x(t)|^2 dt} \quad (1)$$

$$\text{Or : } \text{PAPR}(x(t)) = \frac{\max|x(t)|^2}{E|x(t)|^2} \quad (2)$$

In reality, the probability that the sub-carriers recombine to reach this value is low and in some cases, the PAPR is described using the distribution function which determines the probability that the PAPR exceeds a threshold value, or even its expectation and its standard deviation.

2.2. PAPR distribution function for OFDM; CCDF

The random variable term PAPR comes from the fact that the input symbols arrive randomly at the input of the stage of the modulation. Each symbol is carried by a carrier, and the addition of these carriers can be coherent over certain instants, thus generating amplitude peaks. The analysis of the PAPR may be carried out by a deterministic study which consists of calculating its maximum value or an upper limit. It is also possible that the study be probabilistic using the distribution function CDF (Cumulative Distribution Function) or the complementary function CCDF (Complementary Cumulative Distribution Function). For a Multi-carrier OFDM system, the maximum value of the PAPR is [6]:

$$\text{PAPR}_{\max} = M \quad (3)$$

The probability of having a PAPR equal to the number of carriers for 64 carriers for example is about $2^{-64} \approx 10^{-19}$. It is therefore necessary to use statistical methods to better characterize the PAPR. The CCDF function is defined as being the probability that a random variable exceeds a certain threshold value which is denoted γ by the following relationship indicated in reference [7] and given by:

$$\text{CCDF}(\text{PAPR}) = \Pr(\text{PAPR} \geq \gamma) \quad (4)$$

For large values of M, the CCDF is given by:

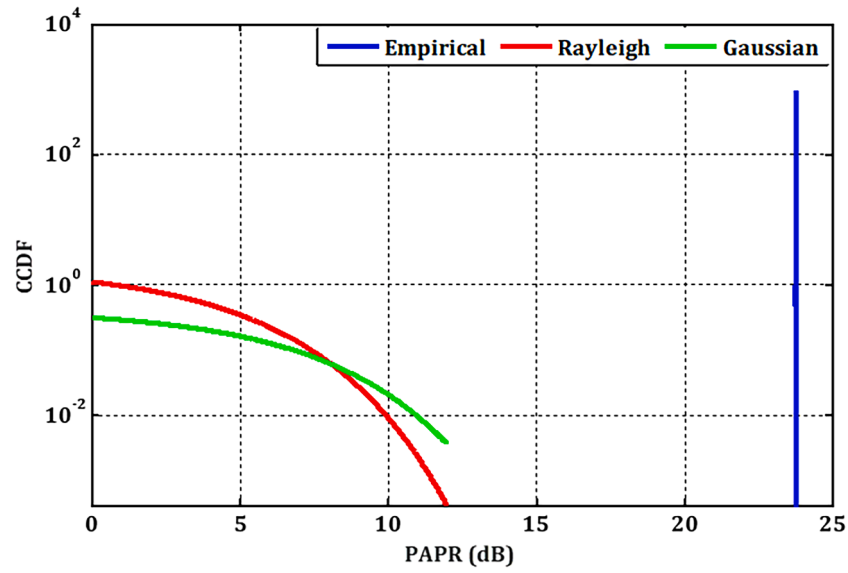


Fig. 2. Comparison between the empirical CCDF and two analytical CCDF approximations based on a Rayleigh distribution and a Gaussian distribution for a 256-carrier OFDM system.

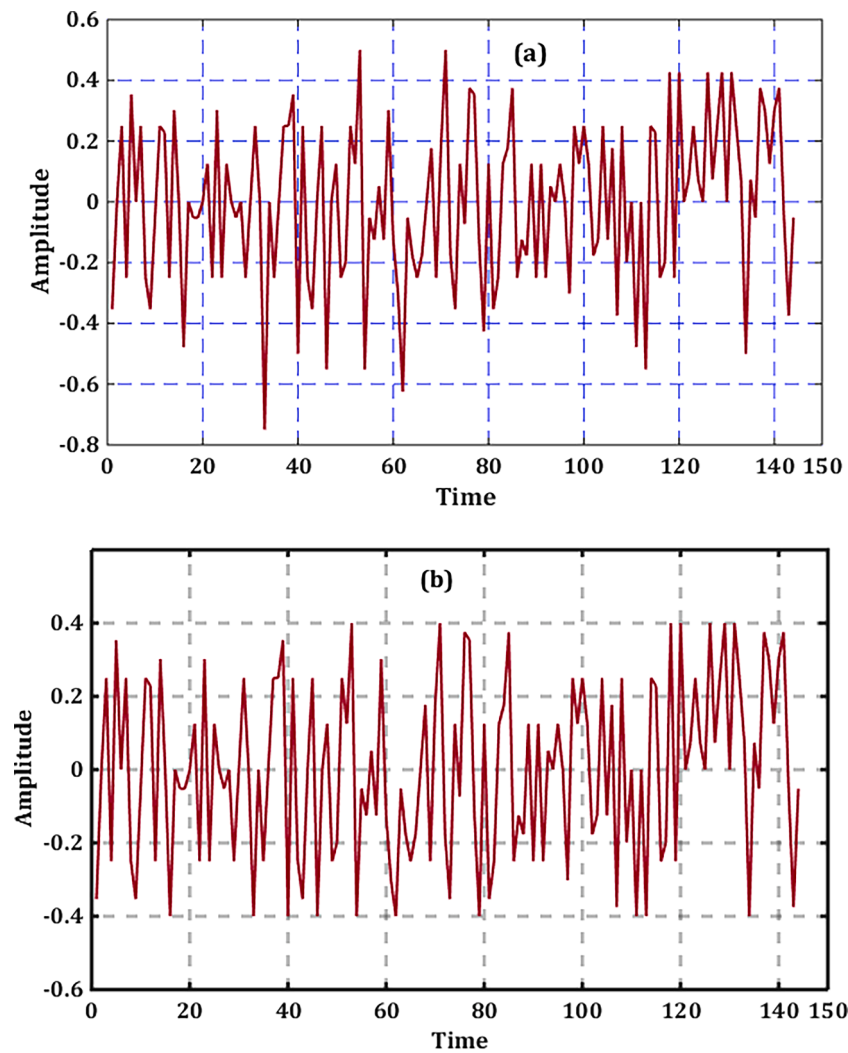


Fig. 3. PAPR reduction by Clipping, a: before clipping, b: after clipping.

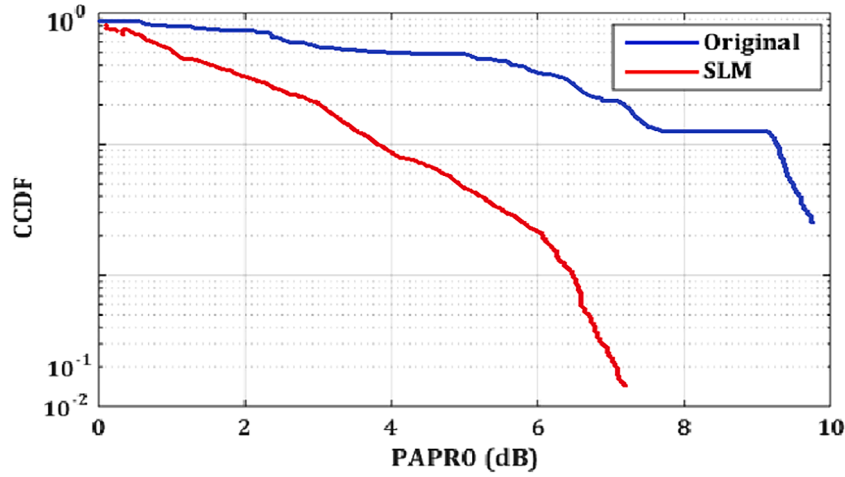


Fig. 4. CCDF for original OFDM and SLM-OFDM.

$$CCDF(PAPR) = \Pr(PAPR \geq \gamma) \approx 1 - (1 - e^{-\gamma})^M \quad (5)$$

However, when oversampling is carried out to model an OFDM signal in the continuous case to make the CCDF measurements closer to those of the analog version, the definition given in the previous equation is no longer valid. An empirical approximation corresponding to the continuous case of the OFDM signal was given by Van Nee and De Wild, modeling the oversampling process as the consideration of a larger number of independent samples between them. This approximation is given by:

$$CCDF(PAPR) = \Pr(PAPR \geq \gamma) \approx 1 - (1 - e^{-\gamma})^{\alpha M} \text{ with } \alpha = 2.8 \quad (6)$$

An oversampling factor of 4 is sufficient to represent the OFDM signal for the continuous case, since for a higher factor, we find the same curve of the CCDF of the PAPR. Another CCDF approximation of OFDM which gives good results from a number of carriers equal to $M = 64$ is given by [8]:

$$\Pr(PAPR \geq \gamma) \leq \sqrt{\frac{\pi}{3}} M \sqrt{-\gamma} e^{-\gamma} \quad (7)$$

More recently, Savaux et al. [9] derived an approximation of the PAPR distribution based on the extreme value theory of the Chi-square process of the OFDM signal. This approach does not assume the uniformity of the power allocation but considers that the power is not allocated in the same way to the different carriers. Let M_{tot} be the total number of carriers, M the number of active carriers, P_m the power allocated to the M^{th} carrier, and P_{Average} the average power of the OFDM signal. An approximation of the PAPR distribution for an OFDM signal is expressed by:

$$\Pr(PAPR > \gamma) \approx 1 - \exp \left\{ -2e^{-\gamma} \sqrt{\frac{\pi\gamma}{M_{\text{tot}}P_{\text{Average}}}} \sum_{m=-\frac{M}{2}}^{\frac{M}{2}} m^2 P_m \right\} \quad (8)$$

By comparing the empirical CCDF with the Rayleigh-based and Gaussian-based analytical CCDF approximations in Fig. 2 we can evaluate the accuracy of the analytical models in representing the PAPR distribution in the specific 256-carrier OFDM system.

The empirical CCDF shows the probability that the PAPR exceeds a certain value given in dB. It can be observed that the PAPR is often less than 10 dB in this system, but that it can exceed 12 dB with very low probability. The Rayleigh distribution-based approximation is often used as a model for PAPR distortions in OFDM systems. It can be observed that this approximation fits the empirical CCDF well for PAPR

values below about 10 dB, but deviates from the empirical CCDF for higher values.

3. Main techniques to reduce the papr of ofdm signals

Different ways of classifying PAPR reduction methods are possible. There are three categories of PAPR reduction methods in the literature [10], namely signal addition techniques, probabilistic techniques, and coding techniques. The state-of-the-art PAPR reduction methods will be briefly presented in this part, with their simulations to keep the best technique from a performance point of view to optimize it later.

3.1. Clipping: the simplest technique which consists of limiting the amplitudes beyond a certain threshold

The clipping technique [11] is simple to implement, but it may be less effective than other PAPR reduction techniques in terms of transmission quality and bit error rate (BER). However, it can be used as a back-up PAPR reduction method when needed, if other techniques are not applicable or not effective enough.

Clipping consists of limiting the amplitude of the OFDM signal below a certain threshold. Parts exceeding this threshold are removed, which reduces the PAPR. The choice of clipping threshold is important to minimize signal distortion while effectively reducing PAPR. Too low a threshold can lead to strong distortion, while too high a threshold can reduce the effectiveness of PAPR reduction. Fig. 3 below illustrates the clipping technique:

In Fig. 3a: before PAPR reduction, the OFDM signal typically exhibits high amplitude peaks. These peaks can lead to inefficient power amplifier utilization and cause undesirable distortions. As a result, the transmission quality of the signal can be compromised. In Fig. 3b: after PAPR reduction, a significant improvement can be observed in the OFDM signal. The excessive amplitude peaks are attenuated through the application of the clipping technique. This enables more efficient utilization of the power amplifier and reduces the risk of distortions.

By reducing the PAPR, the OFDM signal becomes more stable and predictable, facilitating its transmission without significant loss in quality. PAPR reduction also contributes to overall system performance enhancement by increasing transmission capacity and reducing transmission errors. However, it is important to note that PAPR reduction may introduce some distortion to the signal due to clipping. Hence, finding a balance between PAPR reduction and the introduced distortion is crucial to ensure optimal transmission quality.

3.2. Selective mapping (SLM)

The principle is to increase the input symbols resulting from digital modulation by a series of different vectors generating several versions of the OFDM signal at different phases after the IFFT passage.

The version of the signal which represents the best performance in PAPR is retained. This algorithm is complex, especially since it requires performing the IFFT as many times as there are a lot of versions.

This method is based on the use of the optimal PAPR among several [12]. The simplicity of this idea is to multiply the signal by L different P vectors and then pass them through the IFFT. The result with the lowest PAPR is sent.

The main idea behind the SLM (Selected Mapping) technique is to search for the binary sequence that minimizes the PAPR (Peak-to-Average Power Ratio) of the OFDM signal. By applying this binary sequence to the modulated signal, then performing the IFFT and adding the cyclic prefix, we obtain the OFDM signal with reduced PAPR. The process is repeated several times to find the best binary sequence that minimizes the PAPR. The block diagram of the SLM Technique is shown in Fig. 4.

This method requires the receiver to identify the vector P . For this, information must be sent which gives the receiver the value of P . The figure below represents a comparison between an original OFDM signal and a signal to which the SLM technique is applied.

Fig. 4 shows the simulated CCDF curve for the basic OFDM and SLM techniques. It is observed that significant PAPR reduction means the selection of a modulation scheme is also important by which we can reduce a small amount of PAPR. It is clear that for the SLM technique, PAPR reduction is reduced at least by 3 dB compared with basic OFDM.

The evaluation of PAPR is done by calculating the ratio between the maximum peak value of the signal and its effective value (RMS). A lower PAPR indicates that the power of the signal is more evenly distributed, which helps avoid nonlinear distortions in power amplifiers and improves the efficiency of the communication system. By using the SLM technique, it is possible to achieve a significant reduction in PAPR, which can lead to improved performance of the OFDM system. This allows for increased transmission capacity, improved signal quality, and reduced risks of interference and distortions.

However, it should be noted that the SLM technique requires some complexity in terms of calculations and signal processing. Additionally, the effectiveness of PAPR reduction depends on the size of the binary sequence search space and the number of iterations performed. Increasing these parameters can enhance the performance of PAPR reduction but at the expense of complexity and computation time.

In conclusion, the SLM technique is a popular and effective method for reducing PAPR in OFDM signals. It helps improve signal quality, optimize power utilization, and reduce nonlinear distortions. However, it is important to find a balance between processing complexity and desired performance for a specific application. Let $S = [S_1 \dots S_N]$ be the temporal OFDM signal and $l \in [0, L]$

$$P^{(l)} = [P_1^l, P_2^l, \dots, P_N^l] P_i^l = e^{j\Phi_i^l} \quad (9)$$

The l^{th} vector of the set of vectors P . The two vectors S and $P^{(l)}$ have the same number of elements N . After multiplication, we obtain L different vectors:

$$S^l = SP^l, \quad l \in [0, L] \quad (10)$$

The receiver has the set of vectors P , but it must know the correct vector P_1 . For this, the transmitter must send the index l of the vector P . The parameter must be protected against errors by different methods to find the initial vector S . For simplicity, the vector takes the values $(\pm 1, \pm j)$ to avoid complex multiplication. P_i^l

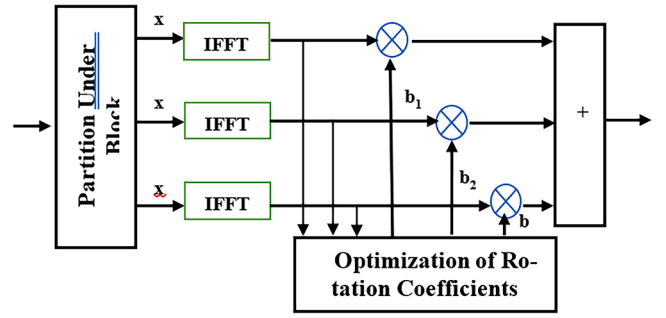


Fig. 5. OFDM transmitter with PTS method

3.3. Partial transmit sequence (PTS)

The principle of the PTS method is very simple and consists of dividing the modulator data vector into M sub-blocks. Then we calculate the PAPR for each sub-block by multiplying the sub-block by a factor b_m , according to the parameter W . We can calculate all the possibilities to have the minimum PAPR. The transmitter must send the information about the combination used to reduce the PAPR to the receiver [10]. Fig. 3 shows the OFDM transmitter with the PTS method.

The technique of reducing the Peak-to-Average Power Ratio (PAPR) of an Orthogonal Frequency Division Multiplexing (OFDM) signal using Partial Transmit Sequence (PTS) is widely employed to address the issue of high PAPR in OFDM communication systems. OFDM is a modulation technique used in various wireless standards such as Wi-Fi, 4 G, and 5 G, offering advantages in terms of interference resilience and channel capacity. The PTS technique is based on dividing the OFDM signal into different partial sequences, which are then transmitted separately with specific amplification. These partial sequences are carefully selected to minimize the overall PAPR of the OFDM signal. Once the sequences are transmitted, they are combined to form the final OFDM signal. The main strength of the PTS technique lies in its ability to effectively reduce PAPR in OFDM systems. By distributing the signal power across different partial sequences, it becomes possible to mitigate high power peaks, leading to improved power amplifier linearity and reduced signal distortions. This results in enhanced signal quality, better energy efficiency, and increased channel capacity (Figs. 5-9 and Table 1).

However, the PTS technique also has certain limitations. Firstly, it requires high computational complexity due to the exhaustive search for optimal combinations of partial sequences. This can result in increased hardware and software complexity in communication systems. Additionally, the introduction of partial sequences and combination mechanisms may introduce some signal delay, which can be detrimental in real-time transmission applications. The figure below illustrates this technique, drawing inspiration from the amplitude versus time curve.

In conclusion, the technique of reducing the PAPR of an OFDM signal using PTS is an effective approach to mitigate the issue of high PAPR in OFDM communication systems. It offers significant advantages in terms of linearity, signal quality, energy efficiency, and channel capacity. However, it requires high computational complexity and may introduce signal delay, necessitating a careful analysis of the trade-offs between system performance and complexity.

3.4. Tone reservation

The reduction of Peak-to-Average Power Ratio (PAPR) is a technique used in wireless communication systems to minimize power variations between transmitted signals. This helps avoid distortions and unwanted interference in the signal. The PAPR reduction curve using the Tone Reservation (TR) technique typically represents the performance of this technique based on the number of tone reservations (or subcarriers) used. TR is a PAPR reduction method based on selectively allocating

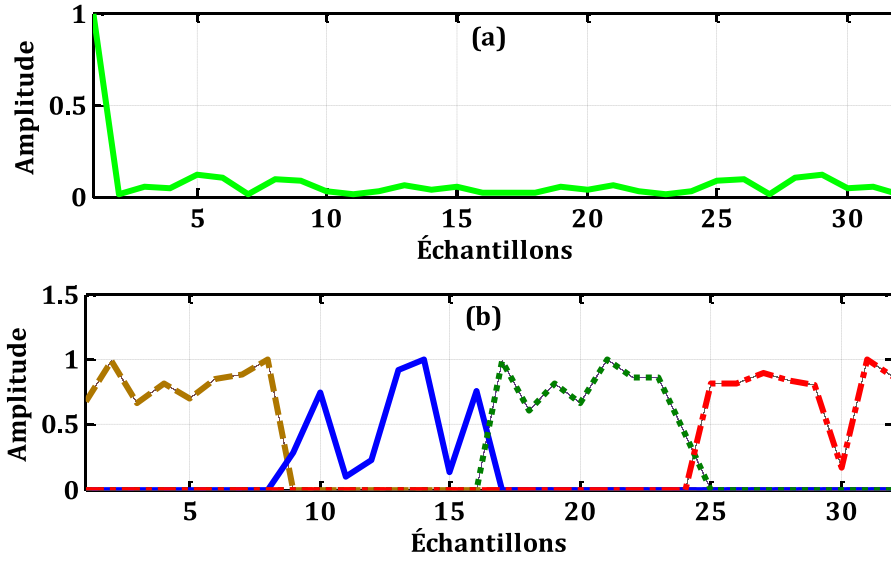


Fig. 6. PAPR reduction by PTS approach: (a) OFDM signal before reduction, (b) OFDM after PTS reduction

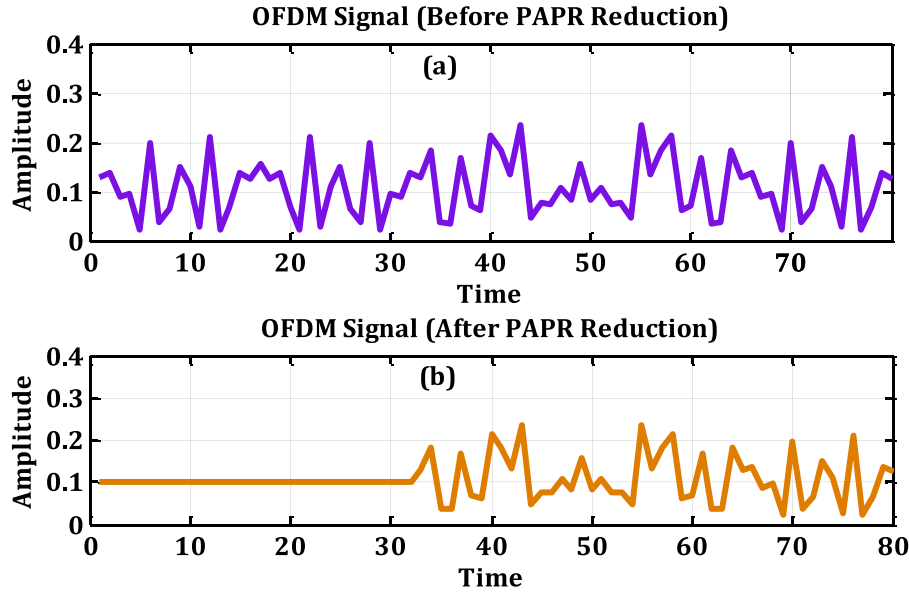


Fig. 7. Tone Reservation technique for PAPR reduction: (a) OFDM signal before reduction, (b) OFDM signal after TR technique

additional tones within the signal's bandwidth [13]. Corrective signal can be achieved through efficient optimization algorithms such as second-order cone programming [14] or the projected gradient (GP) algorithm.

Initially, when the number of tone reservations is low, the PAPR reduction gain may also be low. This means that the TR technique has little significant effect on PAPR reduction. However, as the number of tone reservations increases, the PAPR reduction gain also increases. The curve may exhibit a rapid initial growth trend, where each increase in the number of tone reservations leads to a significant improvement in PAPR reduction. However, as the number of tone reservations continues to increase, the PAPR reduction gain may reach a plateau. This means that adding additional tone reservations only brings marginal improvement in PAPR.

In summary, the PAPR reduction curve using the TR technique shows how the PAPR reduction gain evolves based on the number of tone reservations used. It allows the evaluation of the effectiveness of the TR technique in PAPR reduction and optimizing the trade-off between

system complexity and PAPR reduction performance.

3.5. Criteria for evaluating papr reduction techniques

Given the large number of PAPR reduction methods proposed in the literature [15] performance evaluation is based on several criteria.

- The PAPR reduction gain: This criterion makes it possible to quantify the effectiveness of the technique in terms of the percentage of reduction of the initial PAPR. Its value is estimated at a probability level that the PAPR exceeds a certain threshold. It is generally calculated using CCDF curves (as shown in Fig. 2).
- Backward compatibility: This is a very important criterion, especially in mobile communications. A PAPR reduction technique is called backward compatibility if the receiver is not modified during its installation in the transmitter, such as in the case of clipping, TR, etc. unlike coding techniques which are not downward accounting [16].

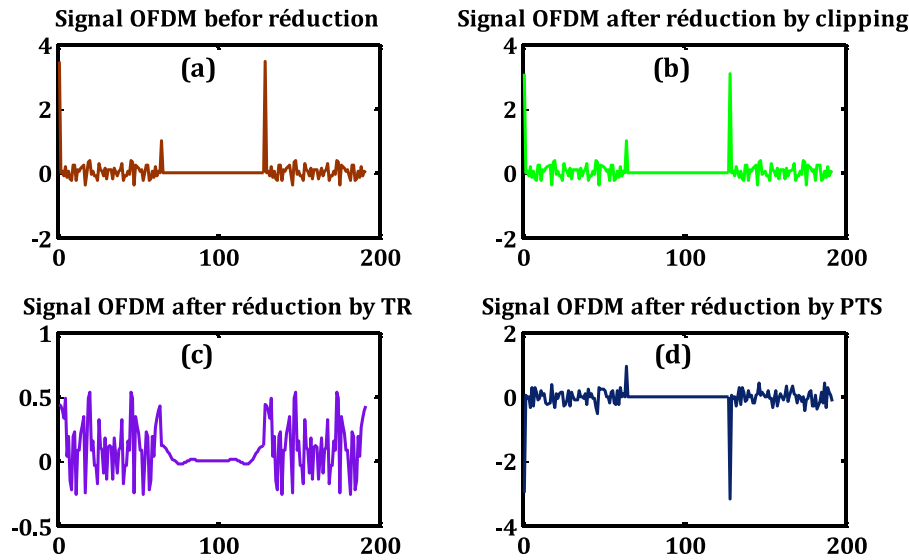


Fig. 8. Comparison of the three PAPR reduction methods: (a) OFDM signal before reduction, (b) signal OFDM after reduction by clipping, (c) OFDM signal after PTS technique, (d) OFDM signal TR technique

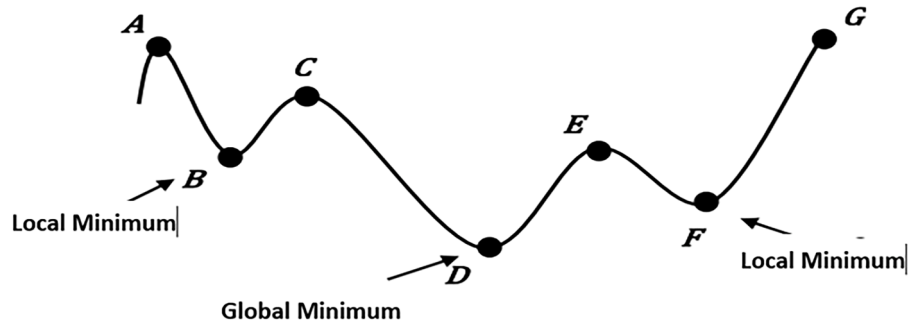


Fig. 9. local and global minimum

Table 1

Results in reduction of PAPR and increase in average power.

	$A = 2.0$	$A = 1.5$	$A = 1.0$
PAPR before TR (dB)	8.61	8.61	8.61
PAPR after TR (dB)	5.45	2.94	4.39
Reduction gain (dB)	3.16	5.67	4.21
Power gain (dB)	0.11	0.91	0.42

- Degradation of the bit error rate (BER): This parameter can contribute to reducing the quality of the communication chain. It can be caused by certain PAPR reduction methods modifying the structure of the signal, introducing distortions. The received signal is then modified and often its BER curve is degraded. • Variation of average power: Techniques with signal addition or peak clipping can increase or decrease the average signal power, which has a strong impact on transmission quality [17].
- Reduction in useful bit rate: The transmitter may be required to transmit information on its PAPR reduction processing to the receiver so that the latter can correctly demodulate the signal. This information reduces the useful throughput of the system.
- Complexity: A technique whose PAPR reduction algorithm is very complex and slow to execute, this will limit its use, even if the PAPR reduction is considerable.

4. Comparison of simulation results and chosen reduction technique

4.1. Comparison of simulation results

From the results obtained in the simulation, we can conclude that the Tone Reservation TR method gives better results than existing ones in terms of PAPR reduction performance and terms of system degradation compared with the three other techniques cited: Clipping, Selective Mapping, and Partial transmitted Sequence PTS.

4.2. Chosen reduction technique: tone reservation (TR) method

4.2.1. Justification of the choice

In this work, we will detail this PAPR reduction technique, indeed the “Tone Reservation” (TR) is by far the most used and is in the process of being implemented in the DVB-T standard because it does not degrade the BER, does not require the sending of edge information (SI) and it is backward compatible. However, the main difficulty of TR lies in the generation of the “PAPR reduction signal” also called the “corrective signal”. It is therefore an optimization problem whose goal is to minimize the level of peaks. This optimization problem can be solved by linear programming:

4.2.2. Definition of the optimization problem

Suppose therefore that the OFDM signal is generated from the data vector X , of size N , and the corrector signal to be added is generated from

the vector C , of the same size N . The vector obtained after application of the method becomes:

$$X_k + C_k = \begin{cases} X_k, & k \in R^{\text{donnees}} \\ C_k, & k \in R^{\text{PRT}} \end{cases} \quad (11)$$

where R^{donnees} represents the set of useful data subcarriers and R^{PRT} is the set of subcarriers dedicated to the correction signal PRT, such that:

$$R^{\text{donnees}} \cap R^{\text{PRT}} = \emptyset \quad (12)$$

The OFDM time signal after the TR method can be described by:

$$(x + c) = Q(X + C) \quad (13)$$

Where $Q = \text{IFFT}$ is the inverse Fourier transform operation allowing the transition from the frequency domain to the time domain in an OFDM modulation defined as follows:

$$Q = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & e^{j\frac{2\pi(1.1)}{N}} & \dots & e^{j\frac{2\pi(1.(N-1))}{N}} \\ 1 & e^{j\frac{2\pi(2.1)}{N}} & \dots & e^{j\frac{2\pi(2.(N-1))}{N}} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j\frac{2\pi(n.1)}{N}} & \dots & e^{j\frac{2\pi(n.(N-1))}{N}} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j\frac{2\pi((N-1).1)}{N}} & \dots & e^{j\frac{2\pi((N-1).(N-1))}{N}} \end{bmatrix} \quad (14)$$

4.2.3. Formulation of the optimization problem

The definition of PAPR given by Eq. (1) $\text{PAPR}(x(t)) = \frac{\max|x(t)|^2}{E|x(t)|^2}$

After applying the Tone Reservation method The previous expression becomes:

$$\text{PAPR} = \frac{\max_k |x_k + c_k|^2}{E\{|x_k + c_k|^2\}} \quad (15)$$

The objective of the TR method is to reduce the PAPR of signal x without increasing its average power, i.e.:

$$\min_c \left| \frac{\max_k |x_k + c_k|^2}{E\{|x_k|^2\}} \right| \quad (16)$$

From the equation, we can formulate the problem of the reduction of the PAPR which results in the following:

$$\min_c \max_k |x_k + c_k|^2 = \min x + c_\infty = \min x + T^{\text{PRT}} \cdot C_\infty \quad (17)$$

Where $\|v\|_\infty$ is the infinite norm of vector v .

In [18], Zabré used the SOCP form (Second Order Cone Programming) to model the convex optimization problem related to the reduction of the PAPR by the TR method. This solution makes it possible to take into account the increase in the average power linked to the method, to the detriment of a high complexity of the algorithm. Therefore it is not suitable for real-time systems. In our study, it is not a question of reviewing the TR method and proposing fundamental changes because its principle is well-known and perfectly mastered. Here, we are rather interested in the use of other algorithms which allow the calculation of the values of the corrective signal. According to the state of the art, no studies are allowed to compare different algorithms and to evaluate the performances in terms of speed of convergence (for the reduction of the PAPR). However, there are studies in the literature comparing the performance of algorithms in terms of PAPR reduction gain, average power increase, or even BER non-degradation.

5. Optimization algorithms

Through this section, we will focus on describing the iterative algorithms that make it possible to solve nonlinear optimization problems. Unconstrained convex optimization algorithms, also known as "direct methods", are mostly based on the general principle of gradient descent [19,20]. These algorithms are mainly used in the search for the optima (generally the minima) of a cost function, also called the criterion to be minimized. From a mathematical point of view, the optimization problem can be posed as follows: let $f(x)$ be a non-linear, continuous, and convex function, where x can be a vector of several elements. Suppose that the shape of the function $f(x)$ can be represented by the figure below, where the Picture example of a shape of the function f .

The optimization problem consists of finding the value of x which gives the minimum (or maximum) value to this function $f(x)$. This can be written as:

Find $\min_x f(x)$, such that $x \in \mathbb{R}^N$

Searching iteratively for the minimum of the function $f(x)$ amounts to going from a point x^i to a point x^{i+1} such that:

$$f(x^{i+1}) < f(x^i) \quad (18)$$

In practice, we start from a point x_0 to reach the optimum by readjusting or correcting at each iteration the value of the vector x according to the formula:

$$x^{i+1} = x^i + d^i \quad (19)$$

With d^i , the direction is chosen for the optimization.

The choice of direction d^i is very important in this case because it is precisely this that differentiates the optimization algorithms.

Now we will describe the following methods:

- The gradient method
- Conjugate gradient methods
- The Quasi-Newton method (also known as the Gauss-Newton method)

Note: Research in the field of parametric identification clearly shows that it is easy to converge to a local optimum. This is essentially linked to the starting point x^0 as well as the choice of search direction. There are several solutions to avoid this problem. The simplest is to take care of the initialization by choosing a nearby starting point of the optimum. This initialization can come from a priori knowledge of the parameters or a series of parametric estimates carried out beforehand

5.1. Gradient algorithm

This algorithm is also known as gradient descent [19]. Its principle is based on an iterative search in a single direction, namely the opposite of the first derivative of the cost function (also called the criterion to be minimized).

After certain iterations, the algorithm converges towards a solution that represents a minimum of the function $f(x)$, also corresponding to a null gradient. We can summarize this iterative search by the following steps:

- Choose the starting point value x_0 .
- At iteration $i + 1$, calculate

$$x^{i+1} = x^i + \mu \cdot \text{sachas} \nabla f_i = -\frac{\partial f}{\partial x} \Big|_{x=x_i} \quad (20)$$

Which is the direction of the search given by the gradient and μ a constant which represents the step of the descent.

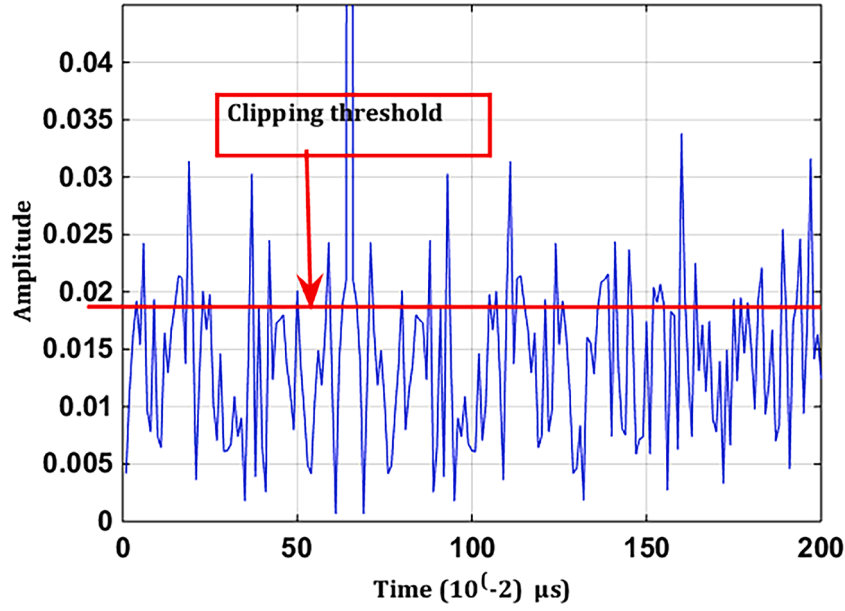


Fig. 10. Example of an OFDM signal with the clipping threshold

- Increment the number of iterations and redo step 2.
- Exit the algorithm when the stopping criterion is verified.

The stopping criterion can be a combination of several conditions. In practice, one imposes a finished number of iterations at the end of which the algorithm is stopped. Also, we can calculate the difference between the costs $f(x^i)$ and $f(x^{i+1})$ and stop the search when this difference is less than a given precision according to the formula:

$$\left| \frac{f^{i+1} - f^i}{f^i} \right| < \text{Precision} \quad (21)$$

The choice of pitch μ is sometimes very tricky. This conditions the convergence of the algorithm. When μ is chosen large enough, the convergence is accelerated at the risk of instability of the algorithm which can diverge. On the other hand, μ is very small, the algorithm will require more iterations but will keep a certain robustness to converge.

5.2. Conjugate gradient algorithm

It is generally used in the field of parametric identification [21]. This makes it possible to reduce the number of iterations with faster convergence. This iterative research can be summarized by the following steps:

1. Choose the starting point value x^0 .
2. At iteration $i + 1$, calculate

$$x^{i+1} = x^i + \lambda^i \cdot d^i \quad (22)$$

With λ^i is the gradient step at iteration i and d^i is the combined research direction given by:

$$d^i = -\nabla f_i + \rho^i \cdot d^{i-1} \quad (23)$$

Increment the number of iterations and do the step 2 again.

1. Exit the algorithm when the stopping criterion is verified.

The conjugate gradient method varies depending on the calculation of the parameter [22] and the Polak-Ribière method [23]. For the

Fletcher-Reeves method, ρ^i can be expressed by:

$$\rho^i = \frac{\nabla f_i^T \nabla f_i}{\nabla f_{i-1}^T \nabla f_{i-1}} \quad (24)$$

While with the Polak-Ribière method, can be given by:

$$\rho^k = \frac{(\nabla f_i - \nabla f_{i-1})^T \cdot \nabla f_i}{\nabla f_{i-1}^T \nabla f_{i-1}} \quad (25)$$

5.3. The quasi-newton method

Generally, the Quasi-Newton method is used to solve nonlinear convex optimization problems. The use of the second derivative called the Hessian allows the number of iterations to be considerably reduced. However, this assumes that the function is differentiable twice, which also limits its use.

The optimal direction search is based on the inverse Hessian [24]. The iterative search based on this algorithm can be described by the following steps:

1. Choose the starting point value x^0
2. At iteration $i + 1$, calculate

$$x^{i+1} = x^i - [H^i + \mu \cdot I]^{-1} \cdot \nabla f_i \quad (26)$$

With $H^i = \frac{\partial^2 f}{\partial x^2} \big|_{x=x^i}$ is the derivative second of the function f which represents the Hessian and I is the identity matrix.

1. Increment the number of iterations and repeat step 2.
2. Exit the algorithm when the stopping criterion is verified.

It can be noted that only the calculation of the increment step differentiates all these algorithms. In the case of the classic gradient, this value is constant. For the conjugate gradient and Quasi-Newton algorithms, the step is variable. For conjugated directions, the search takes into account the previous direction unlike Quasi-Newton, which benefits from the properties of the second derivative of the criterion to be minimized. This last choice is very judicious because it uses the geometric properties of Hessian to readjust the search.

6. Optimization of the tr method

In this section, we will first start with a description of the TR method based on the optimization algorithms seen previously.

6.1. The gradient-based tr method

As we mentioned earlier, the gradient algorithm was first proposed as a solution to the TR method by Kumar [8]. This solution can be implemented in the frequency domain [19] or directly in the time domain. In our work, we will focus on the implementation of the TR method in the frequency domain. From the equation (minimization), we can write:

$$\min_c x + T^{PQR} \cdot C_2^2 \leq A \quad (27)$$

Where $\|v\|_2$ is the Euclidean norm of the vector v and A is a positive real which represents a predefined clipping threshold. We can define a cost function (or the criterion) J to be minimized, such that:

$$J = \frac{1}{2} \sum_{|x_i + c_i| > A} \varepsilon_i^2 = \frac{1}{2} \quad (28)$$

With $\varepsilon_i = |x_i + c_i| - A$ the mean square error between the signal $x + c$ and the predefined clipping threshold A . In this function, we only consider the amplitudes of index i of the signal x which exceed the threshold A (Fig. 10).

In this context, the optimization problem results in the iterative search for the optimal values of the corrective signal c to minimize the quadratic criterion J , and consequently reduce the PAPR of the signal $x + c$. This can be expressed by the following equation:

$$c^{k+1} = c^k - \mu \cdot J'_k \quad (29)$$

c^k is the value of the corrective signal c at iteration k ,

$$J'_k = \left. \frac{\partial J}{\partial c} \right|_{c=c^k} \quad (30)$$

Is the gradient value at point $c=c^k$ and μ represents the gradient step. The analytical expression of the gradient is therefore:

$$J'_k = \sum_{|x_i + c_i| > A} e^{j \cdot \arg(x_i + c_i)} \varepsilon_i^k T_i^{\text{row}} \quad (31)$$

$\arg$ is the phase function and corresponds to line i of the IFFT T matrix. We can therefore summarize this iterative search by the following steps:

1. Initialization step:
 - i. Set the clipping threshold A
 - ii. Set the step value μ
 - iii. Define the set of subcarriers to which the correctivesinal c will be added
 - iv. Initialize c_k to zero and set the maximum number of iterations and/or precision (a stopping criterion).

2. Execution step: This step includes the following operations:

- i. Calculate

$$\varepsilon_i = y_i^k - A \quad (32)$$

Identify only the index amplitude values i of the signal y which exceeded the threshold A .

- i. Calculate the error: $\varepsilon_i = y_i^k - A$; Update correction signal:

$$c^{k+1} = c^k - \mu \cdot \sum_{|x_i + c_i| > A} e^{j \cdot \arg(x_i + c_i)} \varepsilon_i^k T_i^{\text{row}} \quad (33)$$

- ii. Increment the iteration number to $k + 1$ and go to step (2. i) if the stopping criterion is not reached. Otherwise, transmit the signal.

Here, the stopping criterion relates only to the number of iterations because only the speed of the algorithms is evaluated. In practice, we fix the number of iterations to respect a time constraint for emitting the signal. On the other hand, it is difficult to ensure that all the amplitudes that exceed threshold A will be reduced below the latter. Indeed, the optimization algorithm searches for the best compromise to minimize the criterion J (function J) and can, consequently, generate new peaks, during the optimization, in particular because of the constructive sum of the two signals. This problem was addressed in [25] to find the best method to avoid the generation of other peaks by the TR method. Another important point is the initialization of the vector of correction values c_k . In parametric estimation of physical processes, the choice of the initial values of the parameters to be identified is important because it influences the quality of convergence of the algorithm. In practice, we choose to initialize as close as possible to the optimum, which ensures the global minimum for the criterion, to avoid a possible drift towards a local optimum. This initialization can come from a priori information on the parameters (manufacturer data, static tests, etc.) or prior identification. In the case of the TR method, the initialization problem is different because no knowledge about the data to be transmitted is available. Additionally, the PAPR, number, and also location of peaks that exceed threshold A are random and impossible to predict. Finding an initialization common to all symbols therefore seems impossible. Also, we chose to initialize the correction subcarriers to zero. This has two advantages:

No additional calculation is intended to search for initialization for each symbol according to its temporal form,

Above all, maintaining the free subcarriers at zero in the case where the OFDM symbol is below the threshold.

6.2. The tr method based on the conjugate gradient

From the definition of the gradient-conjugate, we can write:

$$C^{k+1} = C^k + \lambda^k \cdot d^k \quad (34)$$

$\text{Or} \lambda^k$ is the conjugate gradient step and d^k represents the conjugate direction of gradient given by:

$$d^k = -J'_k + \rho^k \cdot d^{k-1} \quad (35)$$

We can recall that the calculation of the search direction d^k also considered d^{k-1} For better search efficiency and faster convergence.

As we saw previously, the conjugate gradient method varies depending on the calculation of the parameter that is utilized to build direction. Here we take both versions: Fletcher-Reeves and Polak-Ribière . The Fletcher-Reeves method can be expressed by:

$$\rho^k = \frac{J_k^T \cdot J'_k}{J_{k-1}^T \cdot J'_{k-1}} \quad (36)$$

While with the Polak-Ribière method, can be given by:

$$\rho^k = \frac{(J'_k - J'_{k-1})^T \cdot J'_k}{J_{k-1}^T \cdot J'_{k-1}} \quad (37)$$

6.3. The quasi-newton method

From the description of the Quasi-Newton method given in previous

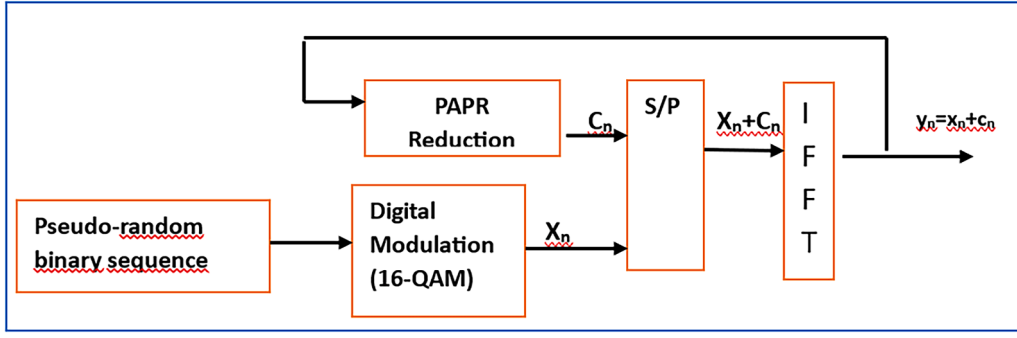


Fig. 11. Simulation scheme of the TR method.

part, we can formulate a new writing for the case of the reduction of the PAPR. Therefore, the PAPR reduction signal be expressed as follows:

$$C^{k+1} = C^k - [J'_k - \mu I_N]^{-1} J'_k \quad (38)$$

Which:

$$J'_k = \left[\frac{\partial^2 J}{\partial^2 C^2} \right]_{C=C^k} \quad (39)$$

is the value of the Hessian of quadratic criterion J and I_N is the matrix identity.

7. Evaluation results of different algorithms

The choice of the optimal number of subcarriers to use to add the corrective signal was made according to studies carried out in the literature. In [26] the author showed that the gain obtained in reduction increases with the number of free subcarriers used. The simulation diagram is presented in Fig. 11.

A pseudo-random sequence is modulated into 16-QAM before performing the IFFT operation for OFDM modulation. At the input of the IFFT module, we have the set of complex data symbols X_n (from modulation) and the set of complex correction symbols C_n . The latter is calculated iteratively by an algorithm optimization represented by the “PAPR reduction” module. The data points and the correction are distributed over all the subcarriers of each OFDM symbol according to the IEEE 802.11a standard. At the output of the IFFT, we have a discrete signal in with a Reduced PAPR, which represents the sum of the two

signals x_n and c_n .

7.1. Optimization of the clipping threshold

We saw that the problem of reducing the PAPR consisted of minimizing a square error between the amplitude of the OFDM signal and a predefined clipping threshold A (Eqs. (28)). We can also deduce that the minimization of this error depends directly on the number of peaks that exceed this threshold. Therefore, the choice of this clipping threshold undoubtedly affects the reduction gain and the execution time of the optimization algorithm.

The following table summarizes the different results obtained in terms of PAPR values and an increase in average power:

- The Reduction Gain is the difference between the PAPR before and after the TR method
- Power Gain reflects the increase in signal power caused by the TR method.

From the results obtained, we can see that the clipping threshold modifies the results in terms of reduction of PAPR and increase of average power.

We can conclude that for a fairly high clipping threshold $A = 2$, the gain in PAPR reduction is approximately 3.16 dB 0.16 dB with a slight increase in average power of approximately 0.11 dB. This is of course expected because a high value of the threshold is easily achievable by optimization but at the expense of lower performance.

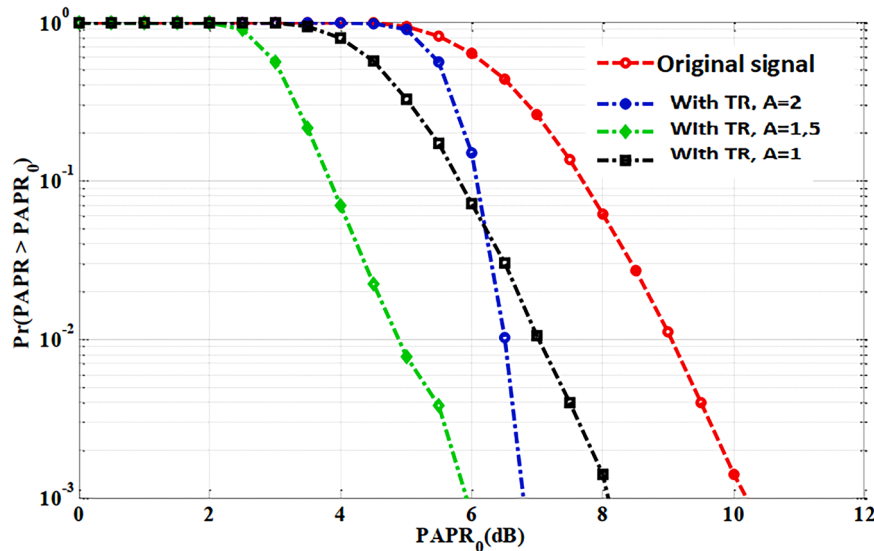


Fig. 12. CCDF of PAPR after 10 iterations.

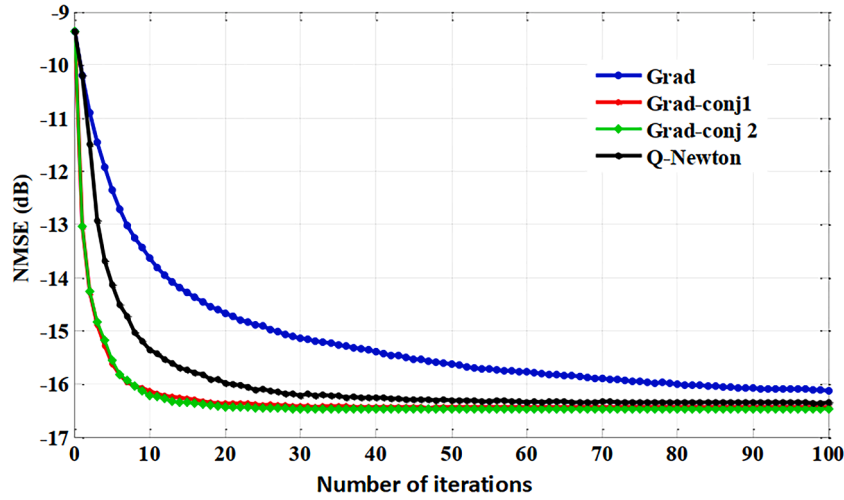


Fig. 13. Evaluation of the convergence speed of the four algorithms studied.

7.2. Impact of the clipping threshold on the CCDF

In (Fig. 12), we have plotted the CCDF for a sequence of 5000 OFDM symbols and each value of the clipping threshold :

In this figure, we can see that choosing a clipping threshold of 1.5 allows us to obtain the best reduction in PAPR, compared to the other two values. This confirms previous results. We can also see that the CCDF curve of threshold $A = 2$ has a different appearance and decreases more quickly than the other curves. This is explained by the perfect clipping of the signal for this value as we concluded. From these results, we can therefore conclude:

- that it is illusory to considerably reduce the threshold in the hope of obtaining a PAPR very low mainly due to constraints linked to the standard,
- that a high value of the threshold allows perfect clipping of the signal but without significant performance improvement.

There is therefore an interval of values for which the reduction of the PAPR is better. In the literature, we do not have precise indications on the choice of the clipping threshold [19]. Theoretically, the search for an optimal threshold requires more information on the distribution of peaks of the OFDM signal.

Therefore, an analytical expression seems to be difficult to obtain.

7.3. Evaluation of the speed of convergence

The speed of convergence of algorithms is a very important criterion for systems evolving in real time. In the case of PAPR reduction by the TR method, it is necessary to have algorithms with rapid convergence. In a real system, the time required to transmit an OFDM symbol is limited by the specifications of the standard. the implementation of the TR method at the transmitter level must respect this time constraint. A simple example of a real-time communication system is the transmission of videos over wireless networks. The objective of this section is to evaluate and compare the speed of convergence of the different algorithms studied previously. To simplify the writing, we will subsequently use the following abbreviations:

- Grad, for the classic Gradient algorithm.
- Grad-Conj1, for the Conjugate Gradient with the Fletcher-Reeves method.
- Grad-Conj2, for the Gradient conjugate with the Polak-Ribière method
- Q-Newton, for the Quasi-Newton algorithm.

We will therefore apply the four previous algorithms on the same sequence composed of 5000 OFDM symbols. At each iteration, we will calculate the normalized mean square error NMSE (Normalized Mean Square Error). The NMSE reflects the value of the quadratic criterion and makes it possible to quantify the difference between threshold A and the peaks that exceed it and is expressed in decibels:

$$NMSE_{dB} = 10 \cdot \log_{10} \left(\sum_{|x_i + c_i| > A} \frac{||x_i + c_i| - A|^2}{A^2} \right) \quad (40)$$

The four algorithms operate under the same conditions, namely a stopping criterion of 100 iterations and a clipping threshold of 1.65.

Fig. 13 shows the evolution of the NMSE as a function of the number of iterations for all algorithms.

The initial value of the NMSE is of course random because it mainly depends on the shape of the OFDM signal studied and the number of peaks exceeding the clipping threshold. As we can see in (Fig. 13), the initial value of NMSE is approximately -9.3 dB. The goal is to reduce this value as quickly as possible, we can also notice that all the algorithms converge towards the same value and allow minimization of NMSE of approximately 7 dB compared to the initial state (from -9.3 dB to -16.3 dB), except the algorithm of the gradient. Indeed, we can notice that this algorithm is longer and does not reach the minimum value after 100 iterations. This result can be explained by the choice of a fixed step of the gradient algorithm which influences the direction of the search. We can also notice that the NMSE cannot be minimized to infinity, but it stabilizes at a certain given threshold. For example, the NMSE is equal to approximately -16.5 Db from the 30th iteration for the conjugate gradient algorithm with its 2 variants. This means that the constructive sum of the OFDM signal to be corrected and the corrective signal to be added by the TR method generates other peaks that affect the NMSE value. In terms of speed of convergence, the conjugate gradient algorithm with its two variants converges more quickly. We can see that the minimum value of NMSE was reached after about ten iterations. In our theoretical study on the algorithms studied, we saw that the Quasi-Newton was faster than the conjugate gradient, which appears to contradict the result that we have just obtained. This is specific to the problem of reducing the PAPR by TR method. Indeed, in a classic optimization problem, the criterion is based on all the samples. In the case of TR, only samples that exceed the threshold are treated in the optimization procedure, which constitutes a special case. Except, for the same symbol, these samples differ from one iteration to another. This affects the quality of the convergence of the Quasi-Newton method because the Hessian calculated for iteration i is found completely modified at iteration $i + 1$.

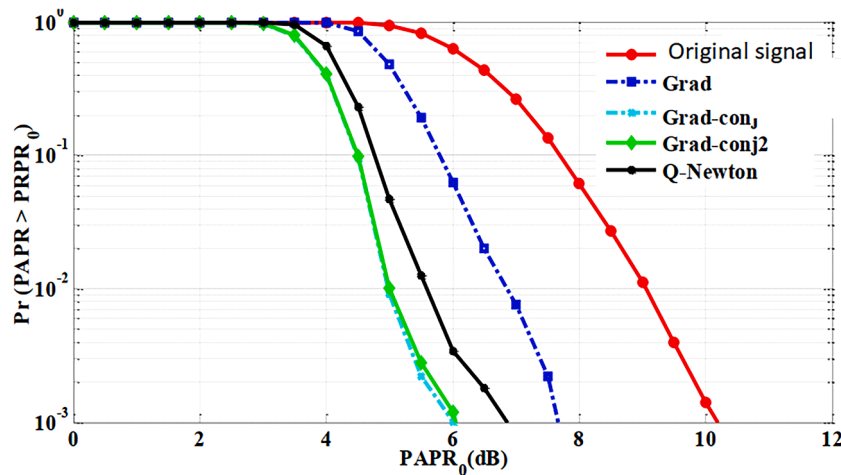


Fig. 14. CCDF of PAPR after 10 iterations.

7.4. Gain in papr reduction

The PAPR reduction gain is defined by the difference between the PAPR of the OFDM signal before and after the application of the TR method. This gain in reduction is a metric important for evaluating the TR method.

In this part, we will see the impact of the algorithms studied on CCDF. (Fig. 14) shows the result for the 4 algorithms studied with stopping the optimization after 10 iterations. We set the number of iterations at 10 to link with the results previously obtained on the NMSE. This makes it possible to translate the minimization of NMSE into a gain in the reduction of the PAPR.

We can notice that all the algorithms make it possible to obtain a reduction in the PAPR after 10 iterations but with different performances. We can also see that the conjugate gradient algorithm with its two variants allows us to obtain better results compared to the simple gradient and Quasi-Newton algorithms. Our objective was to determine an optimal number of iterations to obtain a maximum reduction in PAPR. For this, we compared the simple gradient algorithm (reference algorithm) with the first variant of the conjugate gradient algorithm given that the CCDF curves are superimposed on (Fig. 14) for the two variants, after 5, 10, and 20 iterations. This allows us to see the evolution of the PAPR reduction depending on the number of iterations.

8. Conclusion and perspectives

This work was devoted to the study of PAPR. Therefore, we introduced and briefly analyzed several techniques to overcome and reduce PAPR, including the Tone Reservation technique. This method is widely studied for PAPR reduction in OFDM systems. We introduce an optimization method for TR to enhance the performance of this technique. For that, we compared the performances of the optimization algorithms: The gradient method, The conjugate gradient method, and The Quasi-Newton method (also known as the Gauss-method Newton). We also evaluated the gain in PAPR reduction for the four algorithms,

CRedit authorship contribution statement

Serghini Elaage: . **Abdelmounim Hmamou:** . **Mohammed EL Ghzaoui:** Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Nabil Mrani:** Validation, Supervision, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Conflict of Interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.

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